

CLAE: A Compact 12.5 Hz Bengali Speech Representation Encoder

Anonymous TACL submission

Abstract

We present CLAE, a continuous-latent encoder for Bengali speech that produces 256-dimensional features at 12.5 frames/s with 15.29M parameters. Trained on approximately 2,000 hours of speech, CLAE combines waveform reconstruction, non-contrastive agreement between augmented views, and frame-level distribution regularization. Frozen-feature evaluations show better emotion and speaker discrimination than a random encoder, while specialized pretrained models remain stronger on most tasks. In matched ablations, removing reconstruction degrades reconstruction and temporal-emotion performance; adding view agreement and regularization to reconstruction alone yields mixed gains. Doubling the frame rate improves several metrics, exposing a resolution–performance trade-off. Age prediction does not exceed a majority-label reference. These single-seed comparisons support a compact continuous representation for the evaluated tasks, while leaving generalization, training variability, and perceptual reconstruction quality unresolved.

1 Introduction

Compact speech encoders are useful when their frozen features retain information for several downstream discrimination tasks without requiring task-specific encoder adaptation. This frozen-feature framing follows speech representation work such as wav2vec 2.0 and HuBERT (Baevski et al., 2020; Hsu et al., 2021) and the lightweight-head evaluation protocol of SUPERB (Yang et al., 2021). Modern neural audio codecs and tokenizers also commonly operate at low temporal resolutions, making 12.5 Hz a practical design point for a compact speech representation; DualCodec and NanoCodec, for example, report 12.5 Hz codec

configurations (Li et al., 2025; Casanova et al., 2025). This does not make the systems bitrate- or architecture-equivalent. CLAE is continuous rather than a codec: its floating-point latent is neither quantized nor entropy-coded, so its scalar rate is not directly comparable to codec bitrate.

We present CLAE, a continuous-latent speech autoencoder trained on approximately 2,000 hours from four Bengali speech sources. Its 15.29M-parameter frontend and encoder produce a 256-dimensional frame-level representation z at 12.5 Hz; downstream probes keep this feature extractor frozen and consume z . A distinct 64-dimensional projector output p is used only by two representation objectives: direct global/local view consistency and Variance-Invariance-Sketching Regularization (VISReg; Wu et al., 2026). View consistency encourages features stable to the augmentation and masking nuisances used to construct the views. VISReg regularizes the distribution of projector features to discourage collapse. An auxiliary decoder reconstructs the clean source from a corrupted copy of z , using an 80-bin log-mel loss to encourage information retention.

Our contribution is an empirical study of this combined design for compact Bengali speech representations. We adapt frame-level view agreement and distribution regularization to a low-rate Conformer-derived encoder, assess its frozen features against pretrained speech models, and use matched ablations to examine the objectives, frame rate, decoder corruption, and residual connections. The architectural and objective components build on prior work; the experiments test their combination rather than claim each component as a new method.

2 Related Work

Joint-embedding learning. Image JEPA introduced representation-space prediction with distinct context and target pathways (Assran et al., 2023); A-JEPA adapted masked-target JEPA ideas to audio (Fei et al., 2023). LeJEPA instead studies direct view agreement with isotropic-Gaussian regularization (Balestrieri and LeCun, 2025). CLAE uses a JEPA-inspired, non-contrastive per-frame view-agreement loss, but it is not canonical predictive JEPA: it has one shared encoder and no target encoder, exponential moving average, learned predictor, stop-gradient target path, masked-region prediction, or negative pairs. It also differs from LeJEPA in domain, waveform view construction, grouped consistency loss, frame-level population, supporting regularizer, and reconstruction.

Distribution regularization. VICReg combines invariance, per-dimension variance control, and covariance regularization (Bardes et al., 2022). VISReg retains explicit center and scale terms but replaces covariance control with random one-dimensional projections whose sorted samples are matched to Gaussian quantiles (Wu et al., 2026). We use VISReg to regularize the distribution of projector features and discourage collapse. Our frame-level adaptation is described in Section 3.3.

Continuous and discrete audio representations. Continuous Audio Language Models use a continuous variational bottleneck, KL regularization, additional speech distillation, and generative next-frame modeling (Rouard et al., 2025); those components are not used here. UniWav studies joint representation and generation objectives for speech (Liu et al., 2025a). By contrast, UniAudio and UniTok-Audio illustrate autoregressive generation over discrete audio tokens (Yang et al., 2024; Liu et al., 2025b). Neural codecs such as SoundStream and EnCodec learn quantized, bitrate-controlled representations (Zeghidour et al., 2021; Défossez et al., 2023). Mimi, introduced with Moshi, is a separately pretrained common-evaluation reconstruction reference in our ablation study (Défossez et al., 2024). CLAE’s continuous floating-point latent is neither quantized nor entropy-coded, so its scalar rate is not directly comparable to codec bitrate.

Sequence encoder and residual connections. The encoder builds on the Conformer macaron

block (Gulati et al., 2020). It incorporates squeeze-and-excitation but does not adopt the full downsampling and local-attention design of FastConformer (Rekesh et al., 2023). Selected layers use an adaptation of Manifold-Constrained Hyper-Connections (Xie et al., 2025).

3 Method

3.1 Architecture

Figure 1 summarizes the architecture. For a 16 kHz mono waveform x , the convolutional frontend has five Conv1D stages with channels [128, 256, 384, 512, 512], kernels [10, 8, 8, 4, 4], and strides [5, 4, 4, 4, 4], with GroupNorm and GELU. Its total stride is 1,280 samples, yielding approximately 12.5 frames/s. An eight-layer, 256-dimensional Conformer-derived encoder uses eight attention heads, a 1,024-dimensional feed-forward sublayer, rotary full self-attention, macaron feed-forward modules, a 9-tap convolution branch, dropout 0.1, and squeeze-and-excitation. It produces

$$z = E(F(x)), \quad z \in \mathbb{R}^{T' \times 256}. \quad (1)$$

A per-frame MLP projector with dimensions $256 \rightarrow 512 \rightarrow 64$ produces $p = P(z)$. The decoder and downstream frozen-feature probes use z , whereas the consistency and VISReg objectives use p .

The FiLM-conditioned waveform decoder starts with 768 channels, reverses the frontend with strides [4, 4, 4, 4, 5], and uses two residual blocks per upsampling stage with dilations [1, 3, 9]. The full training model has 23,803,125 parameters. The frontend, encoder, projector, and decoder contain 2,890,240, 12,403,220, 165,440, and 8,344,225 parameters, respectively. The frozen frontend–encoder feature extractor therefore contains 15,293,460 parameters.

3.2 Waveform views and consistency objective

Training first samples one three-second source segment, and every view is derived from that same pre-cropped waveform. Two global views receive independently sampled waveform augmentation. Four local views remain full-length three-second waveforms but receive stronger waveform augmentation and waveform-aligned span masking; their frontend features can receive an additional independent frame mask. The waveform augmentations are gain, additive noise, centered low-pass filtering,

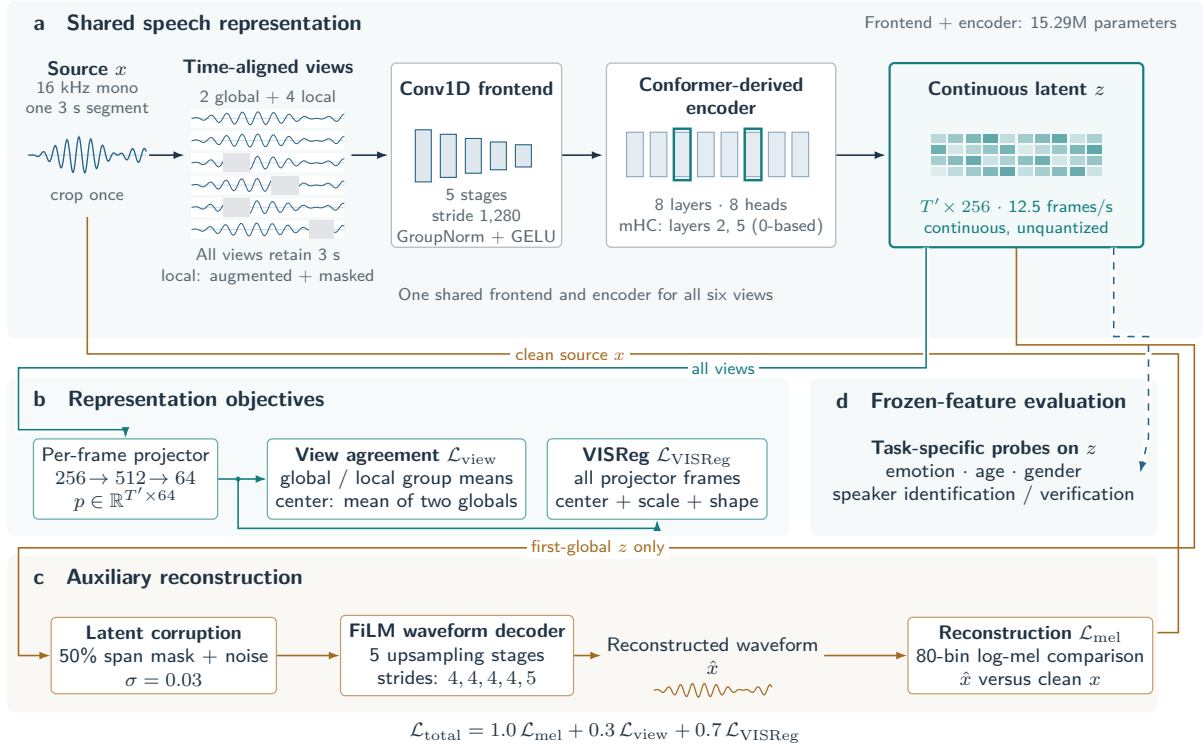


Figure 1: CLAE architecture. (a) Six time-aligned views share the frontend and encoder, producing continuous 256-dimensional features at 12.5 Hz; outlined blocks mark the mHC layers. (b) View agreement and VISReg operate on the 64-dimensional projector output p . (c) The decoder reconstructs the clean source from corrupted first-global features z . (d) Downstream probes use frozen z . Local views receive stronger augmentation and masking but retain the full three-second duration. Waveforms and feature tiles are schematic.

and clipping only: they do not shift, speed-perturb, resample, independently crop, or temporally warp the signal. Waveform masks are constructed on the 1,280-sample frontend-frame grid and expanded onto the corresponding waveform samples; the optional frontend mask is also defined on that frame grid. Thus, frame index t refers to the same time interval across views, although its waveform content may be augmented or masked. All six views pass through the same frontend, encoder, and projector.

Let p_g and p_l denote projector features grouped into global and local views. For each example and frame, the center $\bar{p}_g$ is the mean of the two global projector features. The implemented consistency loss is

$$\mathcal{L}_{\text{view}} = \text{mean}[(p_g - \bar{p}_g)^2] + \text{mean}[(p_l - \bar{p}_g)^2]. \quad (2)$$

Averaging within each group gives global and local views equal aggregate weight despite their different counts. The loss encourages agreement across aligned frames under the specified augmentations. It acts on p without utterance identities or class labels; downstream discrimination is evaluated sep-

arately on the frozen encoder features z .

3.3 VISReg adaptation

Figure 2 shows the frame-level regularizer. We flatten projector features into a population of $B = T'VB_{\text{global}}$ points in $D = 64$ dimensions, where $V = 6$ is the number of views and B_{global} is the source-segment batch size across workers. Frames are gathered across workers without temporal pooling. For population points $q_i \in \mathbb{R}^D$, it penalizes the population mean, centers the points, computes a per-dimension RMS scale with division by $\sqrt{B}$, and penalizes deviation from unit scale. Normalization uses a detached scale. On every forward pass, 256 normalized Gaussian directions are sampled. Values projected onto each direction are sorted and matched to standard-normal quantiles $\Phi^{-1}(i/(B+1))$. Center, scale, and projected-distribution shape terms have equal internal weight.

This implements the core VISReg center/scale/shape construction with CLAE-specific frame-level and distributed adaptations. Our adaptation gathers frames across workers, adds 10^{-6} to the scale for numerical stability, assigns

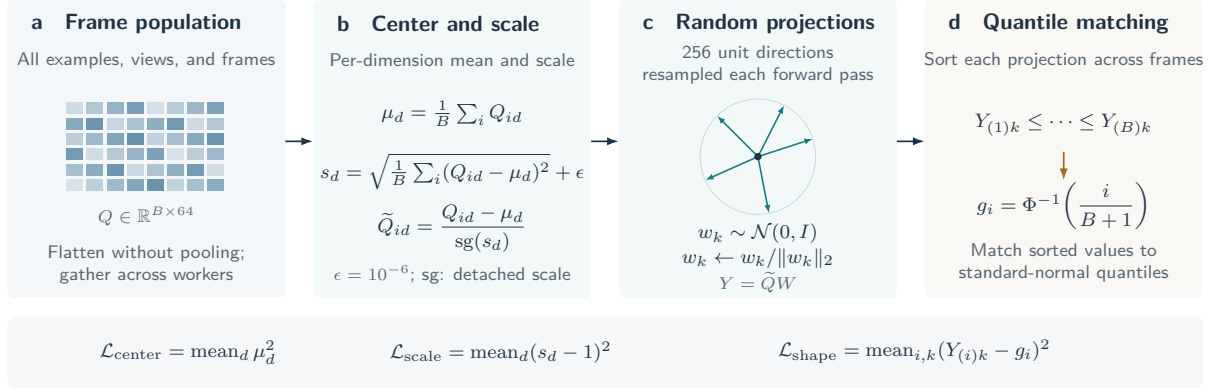


Figure 2: Frame-level VISReg adaptation. (a) All projector frames form one population. (b) Center and scale penalties constrain its first two moments; normalization uses a detached scale. (c–d) Sorted random projections are matched to standard-normal quantiles. The regularizer sums the center, scale, and shape terms with equal internal weights. Tiles and directions are schematic, not measured activations.

equal weights to the three terms, and casts target quantiles to the activation dtype. Training uses BF16 precision.

3.4 Reconstruction and total objective

The decoder receives z from the first global view. Before decoding, an independent span mask removes 50% of frames in spans of 4–24 frames and Gaussian noise with standard deviation 0.03 is added. The target remains the clean source waveform. The decoder outputs a waveform, which is compared with the clean waveform using an 80-bin log-mel loss with FFT size 1,024, window length 1,024, hop length 256, a Hann window, magnitude weight 1.0, log-magnitude weight 1.0, and spectral-convergence weight 0. Multi-resolution STFT is used only for evaluation.

The training objective is

$$\mathcal{L}_{\text{total}} = 1.0\mathcal{L}_{\text{mel}} + 0.3\mathcal{L}_{\text{view}} + 0.7\mathcal{L}_{\text{VISReg}}. \quad (3)$$

The three coefficients weight losses with different numerical scales. Training uses no quantization, variational bottleneck, or adversarial objective.

3.5 mHC adaptation

At the third and sixth encoder layers, mHC widens the residual path to two streams. It applies learned static pre/post mixing and a Sinkhorn-constrained residual matrix with identity blending, uses ten Sinkhorn iterations, wraps complete encoder layers, and returns to one stream by a CLAE-specific uniform mean readout. The resulting residual block is a compact adaptation of mHC. Section 5.2 reports mixed, probe-dependent evidence for this component.

Dataset	Utterances
Common Voice Bengali	1,052,178
OpenSLR-53	218,703
regspeech12	21,313
shrutilipi	17,882
Total	1,310,076

Table 1: Bengali speech sources in the training manifests.

4 Data and Training

The training manifests combine four Bengali speech sources, with 1,310,076 utterances and close to 2,000 total hours of audio: Common Voice (Ardila et al., 2020), OpenSLR-53 (Kjartansson et al., 2018), RegSpeech12 (Hassan et al., 2025), and Shrutilipi (Bhogale et al., 2023). Table 1 gives the source inventory. The manifests use a 95/5 stratified train/validation split with seed 42; Table 2 gives the resulting counts. No exact audited unique duration is available. Common Voice and OpenSLR-53 both contribute to pretraining, so evaluations on those sources are in-domain frozen-feature evaluations with unassessed train–evaluation overlap. The inventory does not establish demographic or dialect balance, and no deduplication or train–evaluation leakage audit was performed. SUBESCO (Sultana et al., 2021), used for emotion evaluation, is absent from the documented training inventory.

We report the terminal checkpoint after 210,000 training steps, without downstream-metric checkpoint selection. Training uses BF16, AdamW, batch size 42, gradient accumulation over four

Split	Utterances
Train	1,244,572
Validation	65,504

Table 2: Stratified 95/5 manifest split (seed 42).

batches, and gradient clipping at 1.0. The learning-rate schedule was adjusted during training: the base learning rate was reduced to 5×10^{-4} at step 50,000, and later continuations used 10^{-4} with a 300,000-step scheduler horizon. Training ended at step 210,000. Thus, the trajectory includes manual schedule changes rather than a single uninterrupted cosine schedule.

With batch size 42 and four accumulation batches on one worker, the nominal effective batch is 168 source segments. If this batch construction is maintained across the trajectory, 210,000 training steps correspond to 35.28M sampled three-second segments, or 29,400 hours of exposure including repeated sampling. These are exposure estimates, not audited counts of unique audio or successful parameter updates. Each source segment produces six augmented views.

There is no active in-loop validation or validation-based early stopping. The terminal checkpoint is reported without a downstream-metric sweep.

5 Evaluation

We assess waveform reconstruction and the information available to lightweight probes trained on frozen encoder features. The full-data protocol evaluates the main pretraining run against external references. A reduced fixed-subset protocol provides the complete six-condition ablation comparison. Their datasets, sample budgets, and probe training budgets differ; scores are compared within each protocol. Appendix A gives the extraction, split, and scoring details.

Reconstruction is evaluated with MR-STFT, PESQ-WB, STOI, and ESTOI (Yamamoto et al., 2020; Rix et al., 2001; Taal et al., 2010; Jensen and Taal, 2016). These objective measures capture complementary spectral and intelligibility properties but do not replace listener judgments. External frozen-feature references are WavLM (Chen et al., 2022), Whisper-tiny (Radford et al., 2023), ECAPA-TDNN (Desplanques et al., 2020), emotion2vec (Ma et al., 2024), Mimi (Défossez et al., 2024), and Higgs Audio V2 (Boson AI, 2025).

Their training objectives, data, and specializations differ. We include an architecture-matched random CLAE encoder to isolate the benefit of pretraining and a training-fold majority-label predictor for the demographic tasks.

5.1 Main pretraining run

Reconstruction uses 400 fixed-order, three-second Common Voice Bengali validation clips with FP32 inference. The reconstruction sample records identify Common Voice 26.0 (June 2026). Generic emotion uses 7,000 SUBESCO utterances with five speaker-disjoint folds. Temporal attention uses 1,708 training and 392 held-out utterances; the temporal Transformer uses four speaker-disjoint folds over 7,000 utterances. Age and gender use five speaker-disjoint Common Voice folds with 15,475 and 15,690 utterances, respectively.

Closed-set speaker identification uses 167 OpenSLR-53 speakers, with 666 training and 334 test utterances. Here the test utterances belong to the same enrolled speakers as the training utterances. Verification instead scores all unique pairs among 2,000 utterances from 334 speakers using cosine similarity. This all-pairs diagnostic is not a standard enrollment/test benchmark. Common Voice and OpenSLR-53 contribute to pretraining, and overlap with these evaluation samples has not been audited.

Table 3 brings the main model and its task-specific references into one comparison. CLAE improves over the random encoder on all five generic representation tasks. Most pretrained references nevertheless have higher emotion and demographic F1, and ECAPA is substantially stronger on speaker identification and verification. Temporal attention and the Transformer measure different probe capacities and use different splits; their scores should not be interpreted as interchangeable estimates of emotion performance.

The age distribution is especially imbalanced: only two clips carry the oldest observed label, and some folds omit rare labels. CLAE’s age macro-F1 is slightly below the majority-label reference. Thus, the demographic experiments demonstrate predictability of some supplied labels, but do not establish useful age prediction beyond that simple reference, demographic fairness, or population coverage. The speaker-identification result likewise supports closed-set discrimination rather than open-set identification.

(a) Waveform reconstruction				
Model	MR-STFT ↓	PESQ-WB ↑	STOI ↑	ESTOI ↑
CLAE (main run)	0.7471	2.146	0.850	0.762
Mimi (8 quantizers)	0.8688	2.007	0.829	0.707

(b) Frozen-feature evaluation (%)							
Features	Linear-probe macro-F1 ↑			Spk-ID acc. ↑	Temporal emotion F1 ↑		Verif. EER ↓
	Emotion	Age	Gender		Attention	Transformer	
CLAE (main run)	48.19	21.58	92.83	65.57	35.89	51.59	27.60
Majority label	—	21.67	43.79	—	—	—	—
Random CLAE	30.49	16.79	74.22	3.89	22.90	28.35	39.05
WavLM	60.18	24.90	93.55	29.64	37.18	61.04	33.89
Whisper-tiny	63.41	23.35	94.38	35.63	—	—	37.47
ECAPA	51.94	25.09	97.26	95.21	—	—	5.99
emotion2vec	62.92	25.94	89.36	32.63	—	—	34.41
Mimi	60.95	25.72	95.70	64.67	—	—	23.08
Higgs Audio V2	45.15	23.50	92.85	72.46	—	—	18.57

Table 3: Main pretraining run and external references under the full-data protocol. Reconstruction averages 400 clips; classification and EER are percentages. Learned probes use seed 0. Temporal probes were evaluated only for CLAE, random CLAE, and WavLM; dashes denote unevaluated task-model pairs. Mimi’s reconstruction uses eight quantizers, whereas its probe features are continuous pre-quantization latents. Verification uses mean+standard-deviation pooling and all-pairs trials.

CLAE improves on Mimi across the four reported reconstruction proxies under this full-data protocol. This comparison concerns reconstruction of common inputs, not equal bitrate or training resources: CLAE emits continuous floating-point features, while the separate Mimi reconstruction reference uses eight quantizers at a nominal 1.1 kbps. The unresolved waveform-level limitation is discussed in Section 6.

5.2 Matched ablations

The ablations retain the same 25,000-step training budget and 100,000-step learning-rate horizon. They compare the full objective with reconstruction only, representation only, removal of mHC, a 25 Hz frame rate, and removal of decoder corruption. The representation-only condition removes reconstruction; reconstruction-only removes both view agreement and VISReg. Therefore, this comparison tests the representation branch jointly and does not isolate VISReg from view agreement. The main pretraining run is included as a longer-trained reference, not as a matched ablation.

The reduced protocol uses 100 reconstruction clips, 1,400 SUBESCO emotion clips, 2,000 Common Voice clips for each demographic task, 500 OpenSLR-53 clips for speaker identification, and 800 OpenSLR-53 clips with 20,000 fixed verification trials. Both temporal-emotion heads use

the same 1,400-clip subset and 20 training epochs. CLAE inference runs in FP32 with autocast disabled. Data, split, and probe seeds are fixed at 0 wherever applicable. ASR is not evaluated in this protocol. Table 4 reports reconstruction and representation results together with the external references on these subsets.

Contribution of the objectives. Representation-only is a negative reconstruction control and trails all other trained CLAE conditions on the reduced representation tasks. This supports retaining reconstruction in the combined design under the evaluated budget, but does not by itself diagnose representation collapse. Conversely, reconstruction-only has the highest generic-emotion F1 among the matched CLAE conditions (43.89%, versus 41.59% for the full objective). The full objective is slightly higher on age and both temporal-emotion probes, while reconstruction-only is higher on gender, speaker identification, and verification. These small, mixed differences do not establish a consistent benefit from adding view agreement and VISReg.

Frame rate and architecture. The 25 Hz condition improves reconstruction proxies, generic emotion, gender, speaker identification, temporal-Transformer F1, and verification relative to the 12.5 Hz full objective, but lowers age and temporal-

(a) Waveform reconstruction							
Model	MR-STFT ↓		PESQ-WB ↑	STOI ↑	ESTOI ↑		
Full objective	0.8679		1.983	0.837	0.730		
Reconstruction only	0.8705		2.052	0.842	0.745		
Representation only	4.2770		1.115	0.323	-0.011		
No mHC	0.8560		2.013	0.835	0.735		
25 Hz	0.8419		2.139	0.865	0.782		
No decoder corruption	0.7888		2.336	0.882	0.805		
Main pretraining run	0.7745		2.167	0.859	0.775		
Mimi (8 quantizers)	0.8949		2.039	0.842	0.723		

(b) Frozen-feature evaluation (%)							
Features	Linear-probe macro-F1 ↑			Spk-ID	Temporal emotion F1 ↑		Verif.
	Emotion	Age	Gender	acc. ↑	Attention	Transformer	EER ↓
Full objective	41.59	26.11	93.23	67.66	35.21	42.30	27.91
Reconstruction only	43.89	25.89	93.32	68.26	34.97	41.13	26.50
Representation only	30.21	21.03	90.59	25.75	17.57	19.13	38.92
No mHC	41.15	24.69	94.13	69.46	30.29	39.55	27.51
25 Hz	43.74	24.77	95.08	70.66	32.41	43.37	23.84
No decoder corruption	43.27	27.19	93.24	73.65	36.72	43.56	30.97
Main pretraining run	43.21	24.16	94.60	65.87	35.59	42.42	27.11
Majority label	—	27.42	44.37	—	—	—	—
Random CLAE	28.85	23.41	67.45	7.78	22.67	26.41	38.33
WavLM	53.36	25.57	94.93	34.13	31.53	47.59	34.19
Whisper-tiny	59.79	27.22	97.16	38.32	—	—	37.93
ECAPA	45.32	30.20	99.09	97.01	—	—	6.32
emotion2vec	55.81	25.12	85.12	36.53	—	—	35.42
Mimi	56.62	30.39	95.85	67.66	—	—	22.85
Higgs Audio V2	42.37	23.80	94.47	74.85	—	—	18.34

Table 4: Matched ablations and external references on the reduced fixed subsets, with data, split, and probe seeds 0. The six ablations use 25,000 training steps; the main pretraining run is shown separately as a longer-trained reference. Reconstruction uses 100 clips; all frozen-feature scores are percentages. Temporal probes use 1,400 emotion clips and 20 epochs. Dashes denote unevaluated combinations. Rows are descriptive results, without significance testing or a composite ranking.

attention F1. Its stronger scores come with twice as many latent frames. Removing mHC also produces mixed results: speaker identification, gender, and verification improve, whereas emotion and age F1 decrease. Neither change supports a uniform advantage across tasks.

Decoder corruption and external references. Removing decoder corruption gives the strongest PESQ, STOI, ESTOI, age, speaker-identification, and temporal-emotion results among CLAE conditions. The longer-trained main model has the lowest MR-STFT. All CLAE age F1 scores remain below the majority-label reference. Whisper-tiny leads generic emotion among the external models, Mimi leads age, and ECAPA leads gender, speaker identification, and verification. WavLM has the strongest evaluated temporal Transformer.

The full-objective CLAE model and Mimi show a metric-dependent reconstruction trade-off on these reduced inputs, rather than the ordering observed under the full-data protocol.

Complementary evidence. An earlier evaluation of the matched ablations used 400 reconstruction clips and larger temporal-emotion datasets (Table 6). It also shows worse performance after removing reconstruction and mixed effects of mHC. That study lacks reconstruction-only results and records a failed no-decoder-corruption reconstruction task; these gaps are not filled with reduced-subset measurements. Table 7 provides additional full-data results, including three-seed neural-probe summaries and the ASR reference. Figure 3 records the matched training trajectories. Training losses describe optimization under each

objective; they are not validation results and do not rank downstream generalization.

6 Limitations

Training did not use validation-based model selection. The main comparisons and matched ablations use one probe seed. Without confidence intervals or independent pretraining replications, small differences cannot establish statistical significance or training stability. No human listening study was conducted, so reconstruction evidence is limited to objective proxy metrics. The saved SI-SDR measurements are unexpectedly low (about -23 to -25 dB for reconstruction-capable CLAE conditions in the reduced study, versus $+2.05$ dB for Mimi). Their disagreement with the spectral and intelligibility proxies remains unresolved; we withhold SI-SDR from the comparison tables pending a waveform-alignment and phase-fidelity investigation. The remaining metrics do not establish sample-level waveform fidelity. We did not compare alternative distribution regularizers under a controlled protocol.

The training inventory was not audited for demographic or dialect coverage, deduplication, or leakage. Common Voice and OpenSLR evaluations are in-domain because both sources contribute to pretraining, and their train–evaluation overlap is unknown. Demographic probes predict dataset-supplied labels rather than independently verified age or gender. Historical corpus versions are incompletely documented, limiting exact replication and interpretation beyond these samples. ASR is a development-set diagnostic, so we draw no conclusion about linguistic retention. The findings concern the described frame-level VISReg and compact mHC adaptations; alternative regularizers and independent pretraining replications remain untested.

7 Ethical Considerations

Speech can identify speakers and reveal sensitive attributes. The age and gender probes measure predictability of corpus labels; they do not validate those labels as a complete account of identity or justify individual-level decisions. Speaker representations can enable unauthorized identification, surveillance, or impersonation. These research evaluations do not establish suitability for deployment in such settings. Bengali speech and acted-emotion results also should not be general-

ized to unrepresented dialects, populations, or spontaneous emotion.

The study uses third-party speech corpora. Their availability does not by itself establish consent for demographic inference or speaker identification. The corpus-specific consent terms, release versions, and redistribution permissions require verification before any audio redistribution; this paper makes no claim that those permissions have been independently audited. No new human listening study or participant recruitment was conducted.

Availability. We plan to release the source code, evaluation recipes, and model checkpoints through a public repository. Dataset manifests will not be released. Third-party audio must be obtained from its providers under the applicable terms. Without released manifests, exact reconstruction of the sampled subsets will be limited; sample counts, label inventories, and split rules are provided for interpreting and replicating the protocols.

8 Conclusion

CLAE learns compact continuous Bengali speech features through reconstruction, view agreement, and frame-level distribution regularization. Its frozen encoder improves emotion and speaker discrimination over a random control, although specialized pretrained references remain stronger on most tasks and age prediction does not exceed the majority baseline. Matched ablations support retaining reconstruction, while the added representation objectives, mHC, and decoder corruption have task-dependent effects. The 25 Hz variant exposes a trade-off between temporal resolution and predictive performance. These findings motivate further study of compact continuous encoders with audited data splits, independent training replications, and perceptual evaluation.

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A Replication Details

CLAE probes use frozen final encoder features z , without layer aggregation or projector features. Audio is mixed to mono and resampled to 16 kHz. CLAE encodes non-overlapping three-second windows, right-pads the final window, and concatenates their frames. Generic utterance probes pool all returned frames, including final-window padding. Reconstruction instead scores the valid waveform prefix when duration is recorded. The two paths therefore have distinct padding conventions.

A.1 Data selection and labels

Demographic subsets are sampled from the first Common Voice validated split in lexicographic file order, retaining entries with available audio, speaker IDs, and nonempty task labels. Eligible entries are shuffled with data seed 0 before the sample cap is applied. Labels are used without merging or correction. The Common Voice release used for these probes is not recorded. Table 5 preserves the raw spelling fourties. Full and reduced age sets contain different classes.

Raw label	Full	Reduced
Age: teens	1,690	197
twenties	12,739	1,674
thirties	871	113
fourties	173	16
fifties	2	0
Gender: female_feminine	3,169	399
male_masculine	12,521	1,601

Table 5: Gold-label counts in the saved out-of-fold predictions. Full/reduced age sets contain 1,580/665 speakers; gender sets contain 1,583/676 speakers.

Generic emotion uses five speaker-group folds on SUBESCO. The full temporal-attention evaluation uses 2,100 clips with 1,708 training and 392 test items; the reduced version uses 1,400 clips with 1,118 training and 282 test items. Both hold out speakers F_09, F_10, M_09, and M_10. The Transformer uses four speaker-group folds. Full speaker ID has 167 enrolled speakers and two held-out utterances per speaker (666 training, 334 test). The reduced set has 500 clips with the same within-speaker holdout rule. These splits separate probe training from testing, not necessarily pretraining from evaluation.

A.2 Probe and metric definitions

Linear probes standardize features using each training fold and fit logistic regression with $C = 1$ and a 3,000-iteration limit; both age and gender use balanced class weights. Speaker ID uses mean pooling; generic emotion, age, and gender use mean+standard-deviation pooling. Reported classification scores are arithmetic means of per-fold scores. Macro-F1 uses labels present in the fold’s gold labels or predictions, as in the evaluator’s default scoring; balanced accuracy averages recall over gold labels present in that fold. Rare-class absence therefore changes the set of scored labels across folds.

The majority-label reference is computed from saved out-of-fold gold labels without feature extraction or model fitting. For each held-out fold, the most frequent label in the other folds is predicted for every test item, and scores are averaged over folds using the same conventions. The majority is twenties for age and male_masculine for gender in every fold. Full/reduced age balanced accuracy is 0.240/0.300 and macro-F1 is 0.216701/0.274245; gender balanced accuracy is 0.500 in both and macro-F1 is 0.437859/0.443730. The same fold-wise convention is used for CLAE and the reference.

Reconstruction runs in FP32 and scores each valid source prefix before averaging over clips. MR-STFT uses FFT sizes [256, 512, 1024, 2048], hop length one quarter of each FFT size, full-length Hann windows, centered analysis, and log-magnitude epsilon 10^{-3} . Spectral-convergence, magnitude, and log-magnitude weights are 0.1, 1, and 1, respectively. STOI and ESTOI use pystoi; wide-band PESQ uses pesq at 16 kHz. Each requested reconstruction metric was computed successfully for all 400 full-data or 100 reduced-subset items. The SI-SDR limitation remains as discussed in the main text.

Verification uses cosine similarity on mean or mean+standard-deviation embeddings; the tables select mean+standard-deviation pooling. Full evaluation scores all unique pairs of 2,000 utterances from 334 speakers. Reduced evaluation has 800 utterances from 134 speakers, with 1,996 target and 18,004 nontarget trials. The capped sampler preferentially retains target pairs before sampling nontarget pairs; it is not uniform all-pairs subsampling. EER averages false-positive and false-negative rates at the nearest ROC crossing. Nor-

malized minDCF uses target prior 0.01 and unit miss/false-alarm costs.

The attention-ASR implementation uses a two-layer, four-head Transformer decoder ($d = 256$, feed-forward dimension 1,024), batch size 16, and an 8,000-update budget. Candidate audio is duration-validated at 15 seconds, encoded in three-second windows, and concatenated without utterance pooling. The full-data ASR development set contains 10,000 items. CER and WER are error ratios, so WER may exceed one. These are fixed-budget development-set diagnostics, not a held-out ASR benchmark.

Temporal probes use a 300-frame cap; the attentive-statistics head has 128 hidden units and batch size 64. The full-data protocol specifies 40 attention epochs and 30 Transformer epochs; the reduced protocol specifies 20 epochs for both. Historical optimizer and stopping settings are incompletely recorded, which limits exact replication. Full neural-probe aggregates use probe seeds 0/1/2 with fixed data/split seeds 0; main and reduced comparisons use probe seed 0. Deterministic linear-probe repeats have identical scalar scores and are not treated as independent training replications.

External feature extractors remain frozen: WavLM-base-plus and Whisper-tiny use final hidden states; WavLM is final-layer only, not a learned weighted layer aggregation. ECAPA uses its utterance embedding; emotion2vec uses continuous FunASR frame features; Mimi uses its continuous pre-quantization latent; and Higgs Audio V2 decodes its quantizer output to continuous vectors. No codec probe substitutes integer code IDs for continuous features. The separate Mimi reconstruction comparator fixes eight quantizers (nominal 1.1 kbps), rather than reusing the pre-quantization probe feature.

A.3 Training configuration and reproducibility

The matched ablations use 25,000 training steps with a 100,000-step scheduler horizon. The training configuration specifies seed 42, AdamW with $\beta = (0.9, 0.999)$, epsilon 10^{-8} , weight decay 10^{-5} , and 5,000 warmup steps. Full-objective loss weights are 1/0.3/0.7 for reconstruction/view/VISReg; reconstruction-only sets the latter two to zero, and representation-only sets the first to zero. No-mHC disables mHC; 25 Hz changes the frontend/decoder stride product to 640;

no-decoder-corruption disables its mask and noise.

The archived experiments do not provide a complete historical software environment, hardware record, or resolved configuration for the main pre-training run. Exact weight identity across the full-data and reduced evaluations has not been independently verified. These gaps limit exact end-to-end replication; the stated training budgets and evaluation protocols should not be read as a complete execution record.

B Complementary Results

Condition	MR-STFT ↓	STOI ↑	ESTOI ↑	PESQ-WB ↑	Attn. F1 ↑	Transf. F1 ↑
Full objective	0.850	0.828	0.715	1.970	34.06	44.15
Representation-only	5.652	0.319	-0.001	1.103	13.17	19.65
No mHC	0.829	0.824	0.721	1.977	28.94	44.89
25 Hz	0.838	0.857	0.767	2.090	31.84	45.82
No decoder corruption	—	—	—	—	31.10	45.05
Mimi, 8 quantizers	0.869	0.829	0.707	2.007	n/a	n/a

Table 6: Earlier matched-budget evaluation: 400 reconstruction clips, 2,100 temporal-attention clips (1,708 train / 392 test), and 7,000 temporal-Transformer clips. CLAE conditions use 25,000 training steps; probes use seed 0. F1 is reported in percent. Reconstruction-only results are unavailable. “Attn.” denotes the temporal-attention probe and “Transf.” the temporal-Transformer probe. Dashes denote the failed reconstruction task in this earlier study. Mimi is separately pretrained, fixes eight quantizers (nominal 1.1 kbps), and is evaluated here only on the common reconstruction inputs.

(a) Reconstruction, neural probes, and verification

Condition	MR-STFT ↓	CER ↓	WER ↓	Attn. F1 ↑	Transf. F1 ↑	EER ↓	minDCF ↓
No decoder corruption	0.775	0.790 ± 0.020	1.106 ± 0.027	30.13 ± 1.08	52.51 ± 0.70	31.91	0.988
Representation only	5.781	0.899 ± 0.019	1.103 ± 0.042	15.93 ± 2.39	22.66 ± 0.53	38.73	0.999
Main pretraining run	0.747	0.795 ± 0.010	1.116 ± 0.014	34.09 ± 1.71	50.77 ± 0.80	27.60	0.988
WavLM (probe seed 0)	—	0.522	0.905	—	—	—	—

(b) Linear probes (%; higher is better)

Condition	Emotion		Age		Gender		Spk-ID
	Acc.	F1	Bal. acc.	F1	Bal. acc.	F1	Acc.
No decoder corruption	48.19	47.77	28.61	22.18	94.75	92.39	71.26
Representation only	35.99	34.96	23.08	15.32	93.87	88.90	16.17
Main pretraining run	48.86	48.19	25.92	21.58	95.43	92.83	65.57

Table 7: Additional full-data results. The first two CLAE conditions use 25,000 training steps; the main run uses 210,000. Neural-probe values are means ± sample SD across probe seeds 0, 1, and 2, with fixed data/split seeds 0. WavLM’s ASR result uses seed 0 only. F1, accuracy, and EER are percentages; CER, WER, and minDCF are ratios. SD measures probe-seed variation, not a confidence interval or variability across pretrained encoders. Verification uses mean+standard-deviation pooling; linear-probe repeats have identical scores. Dashes denote metrics outside the WavLM ASR reference.

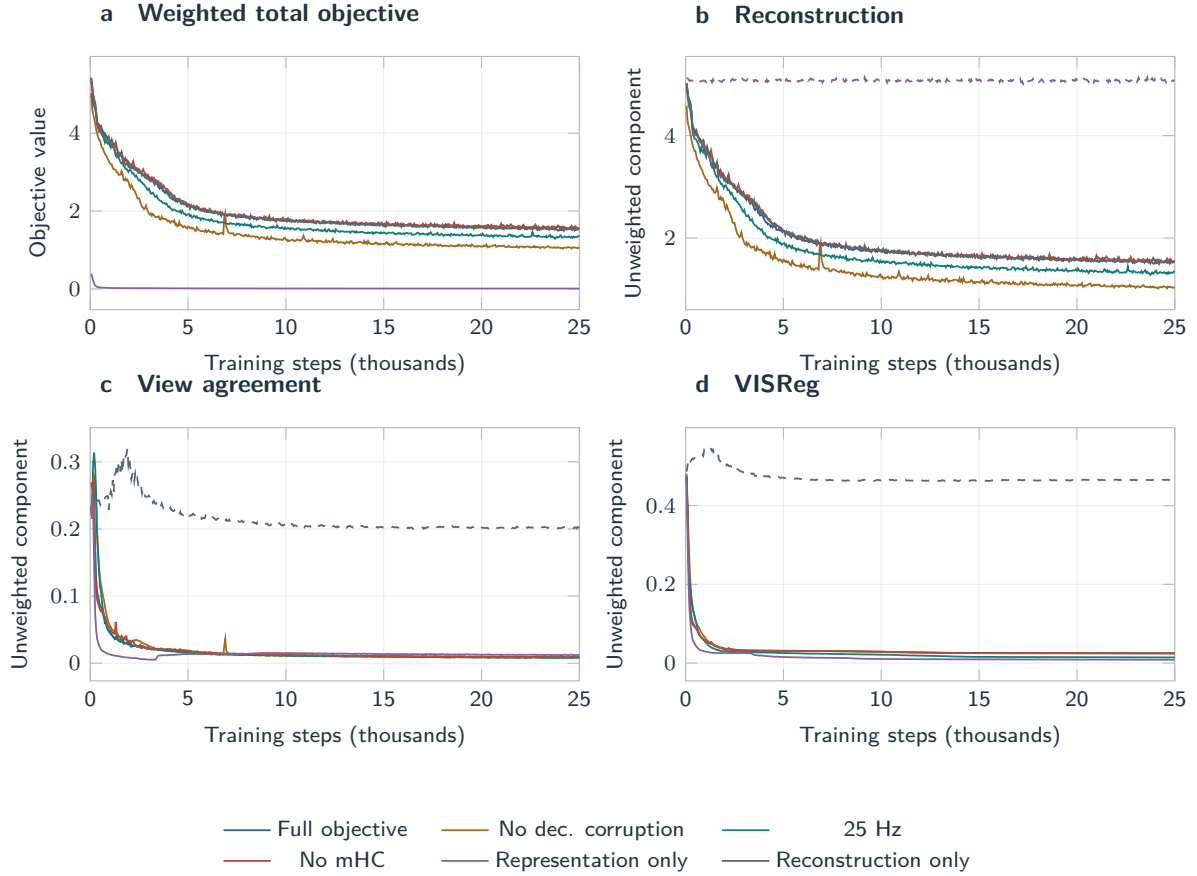


Figure 3: Training trajectories for the six matched ablations. Curves show the original logged values without smoothing. Panel (a) uses each condition’s weighted objective; panels (b)–(d) show unweighted components. Dashed curves denote monitored components excluded from that condition’s objective. The horizontal axis counts attempted training steps, including any skipped updates. Different objectives have different scales, so total loss is not a cross-condition performance ranking. These are optimization diagnostics, not validation curves.